WEEK 4 – FINAL PROJECT SUMMARY REPORT
PROJECT TITLE: SALES DATA ANALYSIS

PROJECT OVERVIEW

This project focuses on analyzing a retail sales dataset to identify trends, patterns, and business insights. The goal was to clean the data, perform exploratory data analysis (EDA), create meaningful visualizations, and build a dashboard to present insights clearly.

The dataset contains information about orders, sales, profit, categories, regions, and order dates.

WHAT I LEARNED

Through this project, I learned:

- How to clean real-world datasets by handling missing values and duplicates.
- How to convert date columns into usable formats (Year, Month extraction).
- How to perform Exploratory Data Analysis (EDA).
- How to create visualizations using Matplotlib and Seaborn.
- How to interpret charts and extract meaningful business insights.
- How to build a simple interactive dashboard in Power BI.
- How to structure a complete data analysis project folder professionally.
- How to document and upload projects to GitHub.

TOOLS USED

The following tools and technologies were used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook
- Power BI (for dashboard)
- GitHub (for project upload)

CHALLENGES FACED

During this project, I faced several challenges:

- Understanding dataset columns and their meanings.
- Handling missing and duplicate values properly.
- Creating clear and readable visualizations.
- Choosing the right type of chart for each analysis.
- Designing a clean and simple dashboard layout.
- Interpreting business insights from numerical data.

These challenges helped me improve my problem-solving and analytical thinking skills.

KEY INSIGHTS FROM THE ANALYSIS

From the sales data analysis, the following insights were discovered:

- The Technology category generated the highest overall sales.
- The West region contributed the highest revenue.
- Some categories had high sales but relatively low profit.
- Sales showed seasonal trends across different months.
- Profit varied significantly across regions and product categories.

These insights can help businesses improve decision-making, optimize product strategy, and focus on high-performing regions.

CONCLUSION

This project provided hands-on experience in complete data analysis workflow from data cleaning to dashboard creation. It improved my understanding of real-world data problems and strengthened my ability to extract actionable insights from data.